Engine Model(s): PT6A

79-20-01

EXTERNAL OIL TUBES - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00 STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire used, unless otherwise specified, shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, 0.025 inch diameter, and will not be specified in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC11-021C	Compound, Cold Carbon Remover
PWC11-027	Solvent, Petroleum
PWC11-031	Cleaner, Engine

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

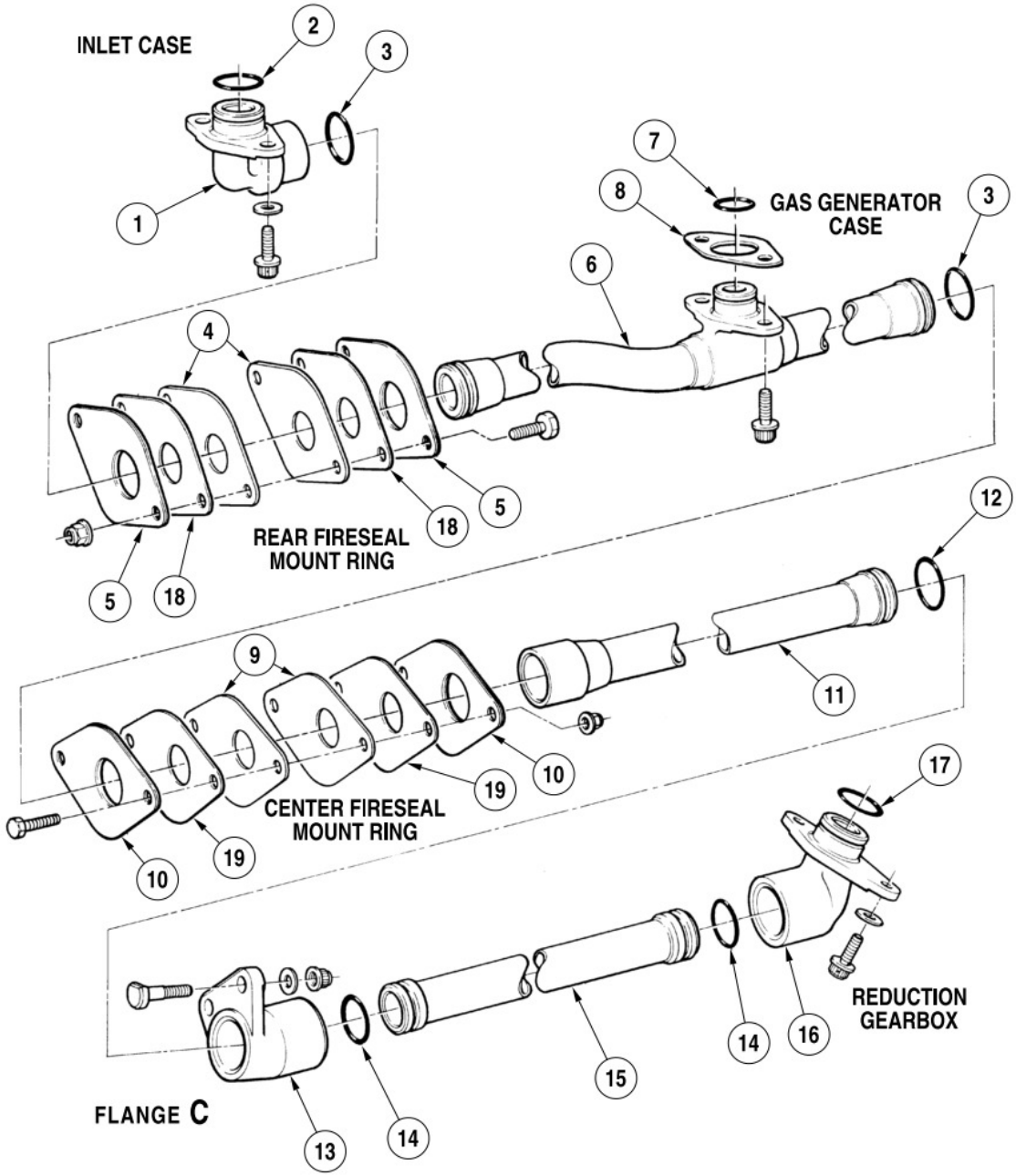
Not Applicable

5. Removal/Installation

A. Removal of Pressure Oil Tube (Ref. Fig. 201)

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Figure 201 Removal/Installation of Pressure Oil Tubes



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1. Elbow (Rear)
2. Preformed Packing
3. Preformed Packing
4. Seal
5. Retaining Plate
6. Rear Tube
7. Preformed Packing
8. Alignment Spacer (if fitted) (Pre-SB1169)
9. Seal
10. Retaining Plate
11. Center Tube (Post-SB1169)
12. Preformed Packing (Post-SB1169)
13. Coupling
14. Preformed Packing
15. Front Tube
16. Elbow (Front)
17. Preformed Packing
18. Insulation
19. Insulation

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NOTE: Early PT6A-27/-28 engines (Pre-SB1169) are fitted with a two-piece pressure oil line requiring alignment spacers for fitment of rear tube. All other engines (all Models) are of Post-SB1169 configuration, having a non-aligned three-piece oil line.

(1) Drain engine oil (Ref. 72-00-00, SERVICING).

(2) For Pre-SB1169 PT6A-27/-28Engines:

(a) Remove self-locking nuts and bolts fastening the retaining plates (5 and 10), insulation (18) (if fitted) and seals (4 and 9) from respective fireseal mount ring.

(b) Remove bolts and washers from elbow (1) from compressor inlet case.

(c) Remove bolts from flange of rear tube (6) from gas generator case.

(d) Simultaneously withdraw elbow from inlet case, and tube (6) from gas generator case.

(e) Remove rear tube (6) by withdrawing from the coupling (13) at flange C and from the center and rear fireseal mount rings, remove the sealing components and alignment spacers (8) (if fitted).

(f) Remove elbow (1) and preformed packings (2, 3 and 7).

(g) Remove bolts, washers and elbow (16) from reduction gearbox.

(h) Remove the front tube (15) by withdrawing the elbow (16) from the reduction gearbox, and the tube from the coupling (13). Remove the elbow and preformed packings (14 and 17).

(3) For PT6A-27/-28 Post-SB1169, and PT6A-21 Engines:

(a) Remove self-locking nuts and bolts fastening the retaining plates (5 and 10), insulation (18) (if fitted) and seals (4 and 9) from respective fireseal mount ring.

(b) Remove bolts and washers from elbow (1) from compressor inlet case.

(c) Remove bolts from flange of rear tube (6) from gas generator case.

(d) Simultaneously withdraw elbow (1) from inlet case, and tube (6) from gas generator case; pull rearward and withdraw the rear tube from the center tube (11).

(e) Remove rear tube (6) from the center and rear fireseal mount rings, remove the sealing components.

(f) Remove elbow (1) and preformed packings (2, 3 and 7).

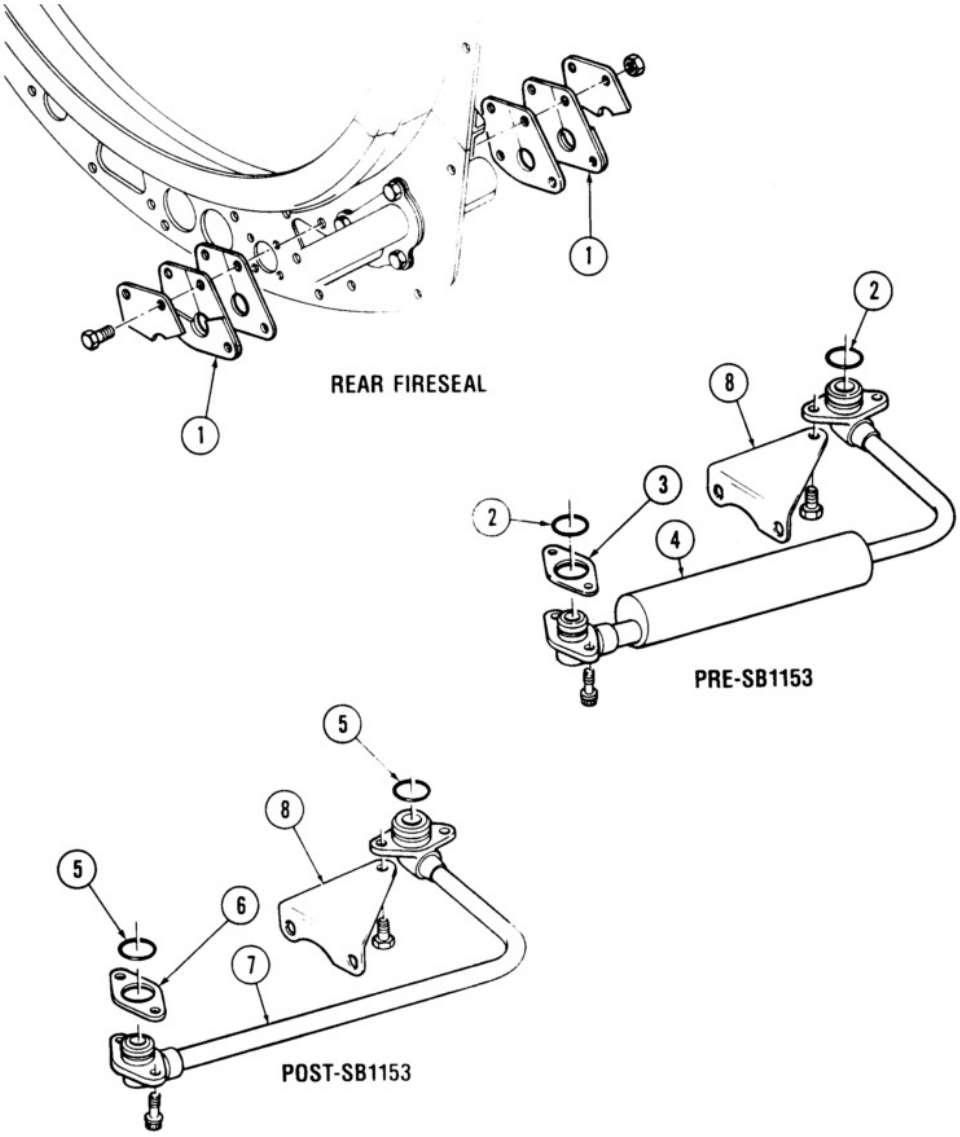
(g) Remove center tube (11) from the coupling (13) at flange C and remove the preformed packing (12).

(h) Remove bolts, washers and elbow (16) from reduction gearbox.

B. Removal of No. 2 Bearing Scavenge Oil Tube (Ref. Fig. 202)

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Figure 202 Removal/Installation of No. 2 Bearing Scavenge Tube



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1. Seals and Seal Retaining Plates
2. Preformed Packing
3. Gasket
4. Tube and Sump Assembly (PT6A-27/-28, Pre-SB1153)
5. Preformed Packing
6. Gasket
7. Oil Scavenge Tube (PT6A-27/-28, Post-SB1153)
8. Rear Fireseal Support Bracket

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(1) Drain engine oil system (Ref. 72-00-00, SERVICING0).

(2) Provide suitable drip pan for residual oil spillage.

(3) Remove self-locking nuts and bolts securing sealing components (1), to rear fireseal mount ring and remove retaining plates.

(4) Remove two bolts securing flange of rear elbow of scavenge tube (7) to boss on gas generator case.

(5) Remove two bolts securing flange of rear elbow of scavenge tube to boss on compressor inlet case.

(6) Withdraw elbows of scavenge tube from respective case and remove tube from rear fireseal mount ring.

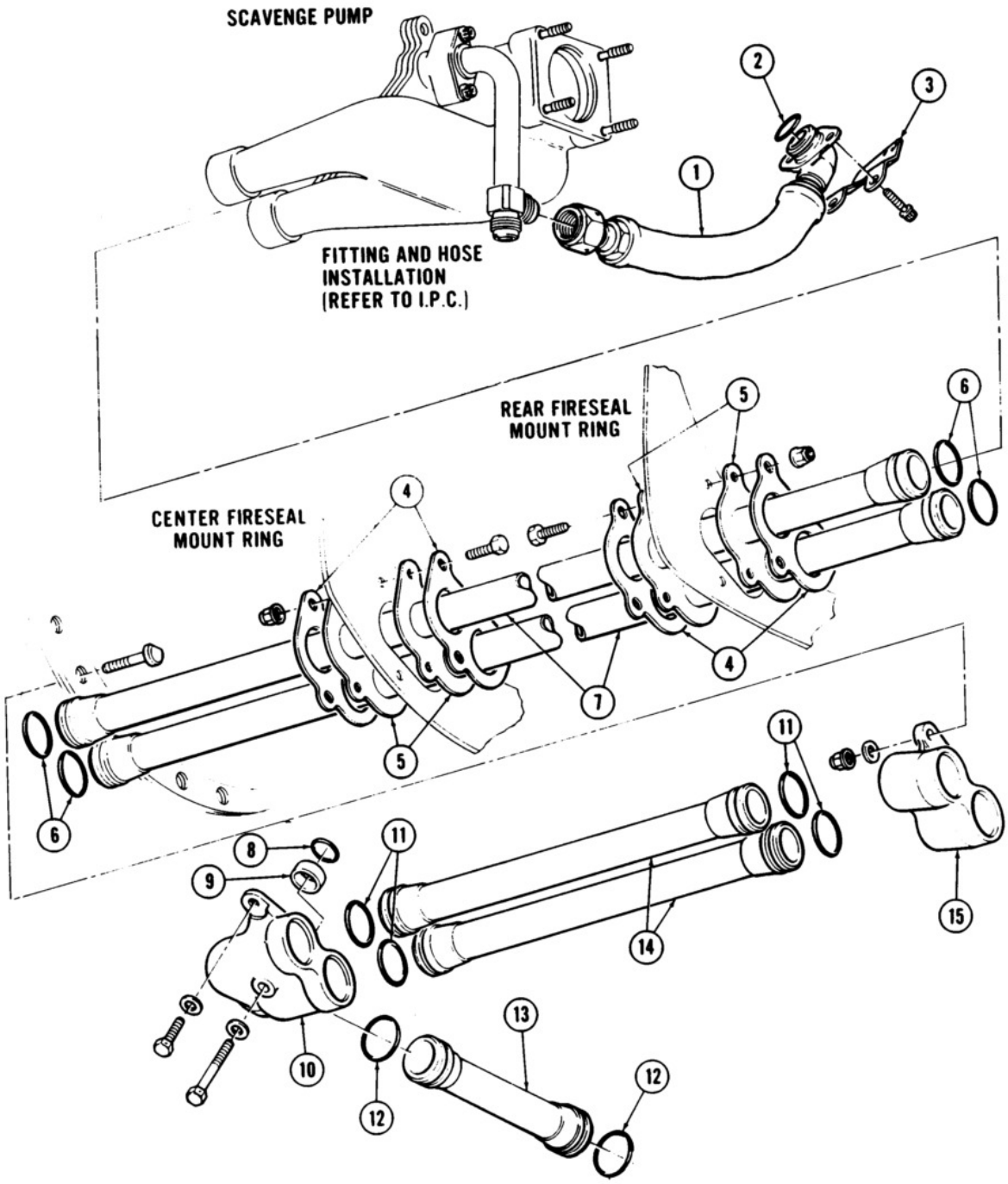
(7) Remove gasket (6), preformed packing (5), and sealing components (1).

C. Removal of External Scavenge Oil Tube (Ref. Fig. 203)

CAUTION: DO NOT SUBJECT EXTERNAL OIL SCAVENGE TUBES TO BENDING OR COMPRESSION STRESSES DURING REMOVAL OR INSTALLATION.

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Figure 203 Removal/Installation of Scavenge Oil Tubes and Hose



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1. Scavenge Oil Hose
2. Preformed Packing
3. Support Bracket (PT6A-27/-28)
4. Retaining Plates
5. Seals
6. Preformed Packing
7. Scavenge Oil Rear Tubes
8. Preformed Packing
9. Sealing Sleeve
10. Dual Elbow
11. Preformed Packings
12. Preformed Packings
13. Transfer Tube
14. Scavenge Oil Front Tubes
15. Dual Coupling

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(1) Drain engine oil system (Ref. 72-00-00, SERVICING).

(2) Provide suitable drip pans for residual oil spillage.

(3) Remove bolts and washers securing dual elbow (10) to propeller reduction gearbox.

(4) Remove Tee-head bolts, washers and self-locking nuts securing dual coupling (15) to flange C.

(5) Remove bolt and self-locking nuts securing seals (5) and retaining plates (4) to center and rear fireseal mount rings.

(6) Withdraw dual elbow (10) from reduction gearbox and remove from transfer tube (13) and front tubes (14). Remove preformed packing (8) and sealing sleeve (9) from elbow.

(7) Withdraw transfer tube (13) from reduction gearbox and remove preformed packings (12) from tube.

(8) Withdraw front tubes (14) from dual coupling (15) and remove preformed packings (11) from tubes.

(9) Remove dual coupling from rear tubes (7).

(10) Remove rear tubes (7) from ports in scavenge pump, and carefully remove tubes from center and rear fireseal mount rings; remove seals and retaining plates during withdrawal of tubes from fireseal mount rings.

(11) Remove preformed packings (6) from rear tubes.

CAUTION: PROVIDE SUITABLE SUPPORT FOR STARTING FLOW CONTROL (PT6A-27/-28) TO PREVENT STRAIN ON REMOVAL OF SUPPORT BRACKET.

(12) Remove two bolts securing support bracket (3) and flanged elbow of scavenge hose (1) to accessory gearbox.

(13) For PT6A-27/-28 Engines: Remove bolt securing support bracket (3) to top of starting flow control. Remove bracket.

(14) Disconnect hose (1) from fitting on scavenge pump; remove hose by withdrawing elbow from accessory gearbox.

NOTE: Configuration of hose depends on actual fitting installed at scavenge pump.

(15) Remove preformed packing (2) from elbow of scavenge hose.

(16) Cap ports in scavenge pump and open boss in reduction gearbox.

D. Installation of Pressure Oil Tube (Ref. Fig. 201)

(1) For Pre-SB1169 PT6A-27/-28 Engines:

(a) Install rear tube (6) with sequential assembly of seals (4), insulation (18) (Ref. IPC) and retaining plates (5) at rear fireseal mount ring, and seals (9), insulation (18) (Ref. IPC) and retaining plates (10) at center fireseal mount ring.

(b) Install preformed packings (3) on each end of the rear tube.

(c) Install preformed packing (2) on elbow (1) and install the elbow on the rear tube.

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(d) Insert the front end of the rear tube (6) into the coupling (13) at flange C and simultaneously insert the flange of the tube into the boss in the gas generator case and elbow (1) into the boss in the inlet case.

(e) Ascertain the quantity of the alignment spacers (8) to be installed between the flange adapter of the rear tube and boss on the gas generator case as follows:

1 Using a feeler gage, measure and record gap between flange of adapter and boss. Make sure a uniform gap exists.

2 Remove the elbow from the air inlet case and withdraw tube from boss and coupling at flange C. Allow tube to rest on fireseals.

3 Install alignment spacers (8), of class determined from measurement obtained in step 1, on the flange followed by the preformed packing (7) on the adapter.

(f) Install rear tube by simultaneously inserting the tube into the coupling (13), the adapter of the tube into the gas generator case and the elbow into the inlet case.

(g) Install the elbow with washers and bolts to the flange with the bolts. Tighten the bolts 26 to 40 lb.in., and fasten with lockwire.

(h) Install the sealing components at the center and rear fireseal mount rings with bolts (heads on the air inlet side of the fireseals) and self-locking nuts. Tighten the nuts until firm contact is obtained, plus 180 degrees.

(i) Install the preformed packings (14) on the front tube (15) and the preformed packing (17) on the elbow (16).

(j) Install the elbow (16) on the front tube (15).

(k) Install the front tube by simultaneously inserting the tube into the coupling (13) at flange C and the elbow into the boss on the reduction gearbox.

(l) Secure the elbow (16) with two washers and bolts. Tighten the bolts 36 to 40 lb.in., and fasten with lockwire.

(m) Replenish the engine lubrication system (Ref. 72-00-00, SERVICING).

(n) Check for oil leaks during engine test run (Ref. 71-00-00, ADJUSTMENT/TEST).

(2) For PT6A-27/-28 Post-SB1169, and PT6A-21 Engines:

(a) Install rear tube (6) with sequential assembly of seals (4), insulation (18) (Ref. IPC) and retaining plates (5) at rear fireseal mount ring, and seals (9), insulation (18) (Ref. IPC) and retaining plates (10) at center fireseal mount ring.

(b) Install preformed packings (3) on each end of the rear tube and preformed packing (7) on the flanged adapter of the tube.

(c) Install preformed packing (12) on the center tube (11), and insert the tube into the coupling (13) at flange C.

(d) Mate the rear tube (6) with the center tube (11) and simultaneously insert the flanged adapter of the rear tube into the boss in the gas generator case and the elbow (1) into the boss in the compressor inlet case.

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(e) Secure the flanged adapter to the gas generator case with two bolts and the elbow to the inlet case with two washers and bolts. Tighten the bolts 36 to 40 lb.in., and fasten with lockwire.

(f) Install the sealing components at the center and rear fireseal mount rings with bolts (heads on the air inlet side of the fireseals) and self-locking nuts. Tighten the nuts until firm contact is obtained, plus 180 degrees.

(g) Install the preformed packings (14) on the front tube (15) and the preformed packing (17) on the elbow (16).

(h) Install the elbow (16) on the front tube (15).

(i) Install the front tube by simultaneously inserting the tube into the coupling (13) at flange C and the elbow into the boss on the reduction gearbox.

(j) Replenish the engine lubrication system (Ref. 72-00-00, SERVICING).

(k) Check for oil leaks during engine test run (Ref. 71-00-00, ADJUSTMENT/TEST).

E. Installation of No. 2 Bearing Scavenge Oil Tube (Ref. Fig. 202)

(1) Pass No. 2 bearing scavenge tube (7) through rear fireseal mount ring with sequential assembly of seals and retaining plate assemblies (1).

(2) Install preformed packings (5) on elbows of scavenge tube.

(3) Locate gasket (6) on front elbow of tube; insert front elbow into boss on gas generator case and rear tube into boss on compressor inlet case.

(4) Secure support bracket and rear elbow to inlet case with two bolts, and front elbow to gas generator case with two bolts. Tighten bolts, torque to 36 to 40 lb.in., and lockwire.

(5) Locate retaining plates on face of respective plate assembly (2), and secure sealing components to rear fireseal mount plate with four bolts (heads to be on air inlet side of fireseal) and self-locking nuts. Tighten nuts until firm contact is obtained plus 180 degrees.

(6) Replenish engine lubrication system (Ref. 72-00-00, SERVICING).

(7) Check for oil leaks during engine test run (Ref. 71-00-00).

F. Installation of External Scavenge Oil Tube (Ref. Fig. 203)

(1) Install preformed packings (6) on each end of rear tubes (7).

(2) Install rear tubes through center and rear fireseals with sequential assembly of seals (5) and retaining plates (4) at each fireseal mount ring. Loosely secure sealing components with three bolts and self-locking nuts at each location.

(3) Insert rear tubes (7) into respective port in scavenge pump.

(4) Locate dual coupling (15) at flange C and mate with rear tubes (7). Secure coupling to flange with three Tee-head bolts, washers and self-locking nuts. Tighten nuts and torque 36 to 40 lb.in.

(5) Install preformed packings (11) on each end of front tubes (14) and preformed packings (12) on transfer tube (13).

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(6) Insert front tubes (14) into dual coupling (15).

(7) Insert transfer tube (13) into port in reduction gearbox.

(8) Insert sealing sleeve (9) into port in mounting face of dual elbow (10) and install preformed packing (8) on spigot.

(9) Install dual elbow by mating with transfer tube (13) and front tubes (14) and simultaneously inserting elbow into boss in reduction gearbox.

(10) Secure dual elbow with three washers, two short bolts and one long bolt. Tighten bolts, torque 36 to 40 lb.in., and lockwire.

(11) Tighten self-locking nuts, at center and rear fireseal mount rings until firm contact is obtained plus 180 degrees.

(12) Install preformed packing (2) on flanged elbow of scavenge hose (1).

(13) Install scavenge hose (1) by inserting elbow into boss in accessory gearbox and connecting hose to fitting on scavenge pump.

NOTE: Configuration of hose depends on actual fitting installed onscavenge pump (Ref. IPC for part and model usage).

(14) Locate support bracket (3) on flange of elbow, and secure bracket and elbow to gearbox with two bolts. Tighten bolts, torque 36 to 40 lb.in., and lockwire.

(15) Tighten coupling nut of scavenge hose to fitting on pump. Torque 450 to 500 lb.in., and lockwire.

(16) Secure starting flow control to support bracket (3) with one bolt. Tighten bolt, torque 36 to 40 lb.in., and lockwire.

(17) Replenish engine oil system (Ref. 72-00-00, SERVICING).

(18) Check for oil leaks at engine test run (Ref. 71-00-00).

6. Cleaning/Painting

A. Cleaning of Tubes

(1) Soak the oil tubes in the cold carbon remover (PWC11-021C) for two hours at room temperature.

(2) Remove tube from solvent and pull a suitable sized swab, or lint-free cloth through tube.

CAUTION: MAKE SURE NO PART OF SWAB OR CLOTH REMAINS IN OIL TUBE; OTHERWISE, OIL FLOW WILL BE RESTRICTED.

(3) Rinse tube exterior and pressure flush tube bore with solvent (PWC11-027) or (PWC11-031).

(4) Dry interior and exterior with dry, compressed air.

7. Inspection/Check

A. Inspection of Tubes

(1) Cracked tubing must be replaced. No repair is permitted.

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(2) Dented tubing must be replaced if any of the following conditions apply:

1. If dent has sharp corners.
2. If dent length and depth exceed 10 percent of tube outside diameter.
3. If more than one dent exists per 12 inch length.
4. If dent is within one inch of ferrule scarf-welds or bend.

(3) Light scratching of no appreciable depth is acceptable. Tubes must be replaced if scratches, nicks, pitting or chafing cannot be smoothed out within 1/2 square inch, to maximum 0.0085 inch deep (Ref. 70-00-00, REPAIR). Any number of such blended areas are permitted provided they are separated by at least 1/2 inch of undamaged material.

(4) Pitting: Tubes must be replaced if pitting cannot be blended out within an area of 1/2 square inch, to a maximum depth of 1/4 of original tube thickness. Any number of such blended areas are permitted, provided they are separated by at least 1/2 inch of undamaged material.

(5) Corrosion: Rust and stains are acceptable if they can be removed by light polishing with fine abrasive cloth, otherwise step (4) applies.

8. Adjustment/Test

Start engine and do checks associated with Installation/Replacement of External Oil Tubes (Ref. 71-00-00, ADJUSTMENT/TEST).

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